

1 (a) (i) Elastic collision (as the total kinetic energy of the system (car and wall) is conserved before and after collision) [B1]

(ii) Considering the toy car and the wall as one system, [B1]

The total momentum is conserved as there is no external resultant force acting on this system. [B1]

There is a corresponding change of momentum to the fixed wall.

Or

Considering the toy car as the system only, [B1]

The momentum is not conserved as there is an external resultant force acting on the toy car by the wall. [B1]

(b) (i) $\Delta p = \text{area under } F - t \text{ graph}$
 $= \frac{1}{2}(0.18)(-15)$
 $= -1.35 \text{ kg m s}^{-1}$ [C1]

$$\begin{aligned}\Delta p &= p_f - p_i \\ &= m(v_f - v_i) \\ &= m(-v - v) \\ -1.35 &= -2mv \\ p_i &= mv \\ &= \frac{1.35}{2} \\ &= 0.675 \text{ kg m s}^{-1}\end{aligned}$$
 [A1]

(ii) $F_{\text{average}} = \frac{1}{2}F_{\text{max}} = \frac{1}{2}(-15) = -7.5 \text{ N}$
or
 $\Delta p = F_{\text{average}} t$
 $-1.35 = F_{\text{average}} (0.18)$ [C1]

$$\begin{aligned}F_{\text{average}} &= -7.5 \text{ N} \\ F_{\text{average}} &= ma_{\text{average}} \\ -7.5 &= 0.65a_{\text{average}} \\ a_{\text{average}} &= -11.5 \text{ m s}^{-2}\end{aligned}$$
 [A1]



Or

$$p_i = mv_i$$

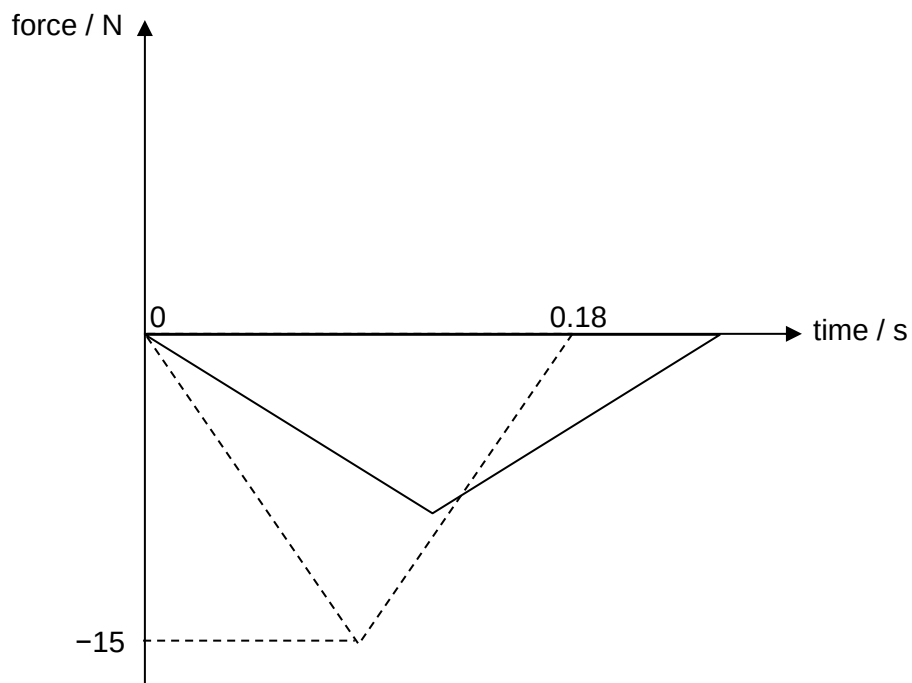
$$v_i = \frac{0.675}{0.65}$$
$$= 1.04 \text{ m s}^{-1}$$

$$a = \frac{v_f - v_i}{t} \quad [\text{C1}]$$

$$= \frac{-1.04 - 1.04}{0.18}$$

$$= -11.6 \text{ m s}^{-2} \quad [\text{A1}]$$

(iii)



[B1] shape with the time increased and magnitude of max force decreased.